

Andrew Matheson

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Education

New York University	New York, NY, USA
<i>Doctor of Philosophy (Neuroscience and Physiology)</i>	2015 – 2021
McGill University	Montreal, Canada
<i>Bachelor of Science (Computer Science and Biology)</i>	2011 – 2015

Research Interests

I am a neuroethologist interested in how animals achieve similar behaviours for survival despite inhabiting vastly different habitats and possessing distinct neural circuit architectures. My current research focuses on the Iberian ribbed newt (*Pleurodeles waltl*), a remarkably plastic species that transitions between aquatic and terrestrial living as an adult — two environments with profoundly different sensory and physical properties. I aim to understand the extent to which nervous systems can be flexible in this species and the morphological, behavioural and genetic modifications newts use to adapt between environments. I plan to leverage the newt's unique plasticity, regenerative capacity and its key evolutionary position as a sister group to all other terrestrial vertebrates to understand the flexibility of the nervous system over evolution. My long term goal is to develop the newt into a full-fledged model for neuroethology and comparative systems neuroscience.

Research Experience

Postdoctoral Research Scientist December 2021 — Present
Advisor: Maria Antonietta Tosches Columbia University, New York, NY

- Studied how Iberian ribbed newts adapt to living between terrestrial and aquatic environments as adults
- Developed a terrestrial husbandry paradigm to induce environmental changes in the lab
- Identified morphological, behavioural and genetic changes between terrestrial and aquatic newts using machine vision algorithms and single-cell transcriptomics
- Developed a novel sensory assay for olfactory navigation in the newt

PhD Student August 2015 — December 2021
Advisor: Katherine Nagel NYU Neuroscience Institute, New York, NY

- Identified wind-direction and appetitive odour pathways from the periphery into the central brain of flies using an optogenetic activation screen and 2-photon calcium imaging
- Determined wind-direction and odour integration occur in the central complex, forming a vector-like representation that could promote olfactory navigation behaviour

Undergraduate Researcher March 2014 — August 2015
Advisor: Jon Sakata McGill University, Montreal, Canada

- Studied the relationship between motor sequencing and timing using songbird behavioural studies

Publications

Matheson AMM, Woych J, Zinga TG, Gattoni G, Saltz VP, Policarpo M, Geiger LT, Spollen N, Jaeger ECB, Gurrola AO, Salzburger W, Tosches MA. Environmental transitions drive plasticity in the body, behavior, and nervous system of a vertebrate, (*in prep*).

Matheson AMM, Chua NJ, Tosches MA, Iberian ribbed newts, *Current Biology*, 2025.

Matheson AMM, Lanz AJ, Currier TA, Licata AM, Syed MH, Nagel KI, A neural circuit for wind-guided olfactory navigation, *Nature Communications*, 2022.

Currier TA, **Matheson AMM**, Nagel KI, A population of central complex neurons promoting airflow-guided orientation in *Drosophila*, *eLife*, 2020.

Suver MP, **Matheson AMM**, Sarkar S, Damiata M, Schoppik D, Nagel KI, Central neurons encoding wind direction in *Drosophila*, *Neuron*, 2019.

Okobi DE*, Banerjee A*, **Matheson AMM**, Phelps SM, Long MA, Cortical control of vocal interaction in neotropical singing mice, *Science*, 2019.†

Matheson AMM, Sakata JT, Relationship between the sequencing and timing of vocal motor elements in birdsong, *PLoS One*, 2015.

*Denotes equal contribution

†Media Coverage: Zimmer, C. These mice sing to one another — politely, New York Times, 2019, at <https://www.nytimes.com/2019/02/28/science/mice-singing-language-brain.html>
 Google Scholar: <https://scholar.google.com/citations?hl=en&user=Tinx-ioAAAAJ>

Oral Presentations

Environmental transitions drive plasticity in the body, behavior, and nervous system of the newt NYU Langone Seminars by Postdocs in Neuroscience: Extramural Series (SPiNES) (New York, NY)	2026
Environmental plasticity in the newt Gordon Research Seminar in Neuroethology: Behaviour, Evolution and Neurobiology (Barga, Italy)	2025
Environmental plasticity in the newt International Society of Neuroethology: Future of Neuroethology Webinar series (virtual)	2025
Environmental brain, body and behavioural plasticity in newts Zuckerman Institute neuroethology club (Zuckerman Institute, NY)	2024
Environmental brain, body and behavioural plasticity in the newt Zuckerman Institute postdoctoral seminar series (Zuckerman Institute, NY)	2024
Uncovering the ancestral circuits for terrestrial vertebrate navigation Janelia Junior Scientist Workshop on Mechanistic Cognitive Neuroscience (Ashburn, VA)	2022
Central circuits supporting olfactory navigation in walking <i>Drosophila</i> Neuromatch3.0 conference, (virtual)	2020
Neural circuits supporting olfactory navigation in walking <i>Drosophila</i> Neuroscience Institute Group Meeting (NYU Neuroscience Institute, NY)	2019
Behavioural-genetic approaches to understanding the neural circuits promoting olfactory navigation NYU Flyclub (NYU Skirball Institute, NY)	2019
Neural populations supporting sensory-motor transformations in <i>Drosophila</i> olfactory navigation FENS-Hertie Winter School: Neural Control of Instinctive and Innate Behaviour (Oberurg, Austria)	2019
Identifying the neural circuits underlying olfactory navigation in <i>Drosophila</i> Neuroscience Institute Group Meeting (NYU Neuroscience Institute, NY)	2018
Identifying the neural circuits underlying olfactory navigation in <i>Drosophila</i> Neurobiology and Behaviour Retreat (McGill University, QC, Canada)	2018
Modelling the control of sequencing and timing in birdsong Neurobiology & Behaviour Seminar Series (McGill University, QC, Canada)	2015

Poster Presentations

Gordon Research Conference in Neuroethology (Barga, Italy) Environmental plasticity in the newt	2025
The Assembly and Function of Neuronal Circuits (Ascona, Switzerland) Environmental brain, body and behavioural plasticity in the newt	2024
Gordon Research Conference in Neuroethology (West Dover, VT) Evolution of terrestrial navigation: insights from the salamander brain	2023
Neuro-Evo: A Comparative Approach to Cracking Circuit Function III (Janelia research campus, VA) Evolution of terrestrial navigation: insights from the salamander brain	2023
Cold Spring Harbor Laboratory: <i>Drosophila</i> Neurobiology (Virtual) A neural circuit for flexible wind-guided olfactory navigation	2021
Cosyne (Virtual) A circuit basis for flexible orientation to wind	2021
Neuronex3 (Virtual) Neural populations supporting olfactory navigation in walking <i>Drosophila</i>	2020
International Symposium on Olfaction and Taste (Virtual) Neural Populations Supporting Olfactory Navigation in Walking <i>Drosophila</i>	2020
Cold Spring Harbor Laboratory: <i>Drosophila</i> Neurobiology (Cold Spring Harbor, NY) Neural populations supporting olfactory navigation in walking <i>Drosophila</i>	2019
Gordon Research Conference in Neuroethology (West Dover, VT) Neural populations supporting olfactory navigation in walking <i>Drosophila</i>	2019
FENS-Hertie Winter School: Neural Control of Instinctive and Innate Behaviour (Oberurg, Austria) Neural populations supporting sensory-motor transformations in <i>Drosophila</i> olfactory navigation	2019

NSF Olfactory Ideas Lab (Washington, DC)	2017
From behavioural algorithms to neural circuits: olfactory navigation in <i>Drosophila</i>	

Practical Courses and Workshops

Junior Scientist Workshop on Mechanistic Cognitive Neuroscience	2022
Janelia Research Campus	
Science Communication Workshop	2019
Arthur L. Carter Journalism Institute, New York University	
Neural Control of Instinctive and Innate Behaviour	2019
FENS-Hertie Winter School, Obergurgl, Austria	

Mentorship

Mohammad Keshktar, <i>Boston University undergraduate student</i>	Summer 2018
Aaron Lanz*, <i>NYU Neuroscience PhD rotation student</i>	Spring — Summer 2020
Dreyton Amador, <i>Columbia University undergraduate student</i>	Summer 2022
Bella Succi, <i>Columbia University Biological Sciences PhD rotation student</i>	Winter 2023
Lekha Masoudi, <i>Columbia University Biological Sciences undergraduate student</i>	Summer 2023
Luke Geiger*, <i>Columbia University Biological Sciences PhD rotation student</i>	Fall 2023
Natalie Spollen*, <i>Columbia University Engineering undergraduate student</i>	Spring 2024
Johanna de la Cruz, <i>Columbia University Biological Sciences PhD rotation student</i>	Winter 2025
Gabriel Graham, <i>Columbia University Neurobiology and Behaviour PhD rotation student</i>	Spring 2026
Therese Grace Zinga*, <i>Columbia University Engineering undergraduate student</i>	Fall 2024 — Present

*resulted/will result in co-authorship on a publication

Awards and Honours

Allison J. Doupe Fellowship, McKnight Endowment Fund	2026
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Teaching

Guest Lecturer, Neurobiology II — Developmental and Systems Neuroscience, Columbia University	2026
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Outreach and Service

Amphibian and Reptile Satellite Meeting Planning Committee	2025 — Present
Planning committee for a subgroup meeting on day before the International Congress of Neuroethology in Vancouver, Canada. Organized abstract submission, coordinated with university vendors, designed logos, and typeset the abstract booklet. Only trainee member of the committee.	
BUMP Biology	2024 — Present
Mentorship Program in Biology: Provide career and general one-on-one mentorship to students interested in a career in biological sciences research. One student has also developed her own research project within the lab that I supervise (Therese Grace Zinga).	
NYU Neuroscience Application Support Group	2020 — 2022
This group paired prospective and current students for mentorship and feedback on their graduate school applications in the form of video calls and written feedback on research statements and CVs. I mentored 1-2 students per year. The majority of prospective students who participated in this program identified as coming from underrepresented backgrounds in science and data showed participants were more likely than average to receive an interview and acceptance.	
NeuroPIL (Neuroscience Predoctoral Interest League) Student Affairs Committee	2020 — 2021
Contributed to guides for incoming students, specifically on international student issues.	
Senior Neurobiology Seminar Guest, Colby College	2020
Senior research seminar guest to discuss a recent publication and discuss careers in academia.	
NYU Brain Day Research Fair, Model Organism Station	2016 — 2019
Science communication and outreach to the local community at NYU Langone. Communicated the use of invertebrate models in neuroscience and ran model experiments as demonstrations.	